Intersection of Two Lines in R^2 and R^3

3. Determine whether the following pairs of lines are coincident, parallel and distinct, or neither.

a. $\frac{x-3}{10} = \frac{y-8}{-4}$

$\frac{x-33}{-5} = \frac{y+4}{2}$

b. $x = 6 - 18s, y = 12 + 3s$

$x = 8 - 6t, y = 4 + 9t$

c. $x = 8 + 12s, y = 4 - 4s, z = 3 - 6s$

$x = 2 - 4t, y = 2 + t, z = 6 + 2t$

d. $\frac{x+4}{3} = \frac{y-12}{4} = \frac{z-3}{6}$

$\frac{x}{\frac{1}{2}} = \frac{y-10}{\frac{2}{3}} = z + 5$

Part B

4. Find the intersection of each pair of lines. If they do not meet, determine whether they are parallel and distinct or skew.

a. $\vec{r} = (-2, 0, -3) + t(5, 1, 3)$

$\vec{r} = (5, 8, -6) + u(-1, 2, -3)$

b. $x = 1 + t, y = 1 + 2t, z = 1 - 3t$

$x = 3 - 2u, y = 5 - 4u, z = -5 + 6u$

c. $\vec{r} = (2, -1, 0) + t(1, 2, -3)$

$\vec{r} = (-1, 1, 2) + u(-2, 1, 1)$

d. $(x, y, z) = (1 + t, 2 + t, -t)$

$(x, y, z) = (3 - 2u, 4 - 2u, -1 + 2u)$

e. $\frac{x-3}{4} = y - 2 = z - 2$

$\frac{x-2}{-3} = \frac{y+1}{2} = \frac{z-2}{-1}$

5. Consider the lines $\vec{r} = (1, -1, 1) + t(3, 2, 1)$ and $\vec{r} = (-2, -3, 0) + u(1, 2, 3)$.

a. Find their point of intersection.

b. Find a vector equation for the line perpendicular to both of the given lines that passes through their point of intersection.

6. Show that the lines $\vec{r} = (4, 7, -1) + t(4, 8, -4)$ and $\vec{r} = (1, 5, 4) + u(-1, 2, 3)$ intersect at right angles and find the point of intersection.

7. If they exist, find the x-, y-, and z-intercepts of the line $x = 24 + 7t$, $y = 4 + t$, $z = -20 - 5t$.

Application

8. Find the point at which the normal through the point $(3, -4)$ to the line $10x + 4y - 101 = 0$ intersects the line.

11. Find the equation of the line through the point $(-5, -4, 2)$ that intersects the line at $\vec{r} = (7, -13, 8) + t(1, 2, -2)$ at 90°. Determine the point of intersection.

Application

12. Find the points of intersection of the line $\vec{r} = (0, 5, 3) + t(1, -3, -2)$ with the sphere $x^2 + y^2 + z^2 = 6$. Is the segment of the line between the intersection points a diameter of the sphere?

13. Find a vector equation for the line through the origin that intersects both of the lines $\vec{r} = (2, -16, 19) + t(1, 1, -4)$ and $\vec{r} = (14, 19, -2) + u(-2, 1, 2)$.

Answers:

2. a. $(-5, -1)$ **b.** $(1, -2)$ **3. a.** coincident **b.** neither **c.** neither **d.** parallel and distinct **4. a.** $(8, 2, 3)$ **b.** lines are coincident **c.** skew **d.** parallel and distinct **e.** $(-1, 1, 1)$ **5. a.** $(-2, -3, 0)$ **b.** $\vec{r} = (-2, -3, 0) + s(1, -2, 1)$ **6.** $(2, 3, 1)$ **7.** x intercept is -4 **8.** $\left(\frac{21}{2}, -1\right)$ **11.** $\vec{r} = (-5, -4, 2) + t(14, -5, 2)$; $(9, -9, 4)$ **12.** $(2, -1, -1)$, $(1, 2, 1)$, No **13.** $\vec{r} = s(17, -15, -20)$ **14. a.** $\left(\frac{-AC}{A^2 + B^2}, \frac{-BC}{A^2 + B^2}\right)$